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Curriculum Assessment of Higher Educational Institution Using Aggregate Profile Clustering

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Abstract

Curriculum assessment commonly takes place on the success story of the students during the education period using some ways such as questionnaires, interviews, etc. The existing approaches frequently used are inadequate to measure the effectiveness of the curriculum as the educational process guideline. Curriculum mining aimed at investigating the actual curriculum model by constructing the model from the students' academic results. However, the discrepancy measurement between the observed behavior of students and the standard curriculum model remains some challenges. This study proposes a methodology to assess the curriculum based on the students' behavior. First, aggregate profile clustering is used to categorize the students according to their learning paths. Second, sequence mining approach is applied to assess the sequence of learning path in conformity with the prior curriculum guideline. The study utilized Information Systems (IS) students' data to test the effectivity of the methodology. Results from the implementation shows that the IS students can be grouped into three clusters that have different characteristics with respect to their performance and learning behavior. It is also found that students with relatively high GPA tend to take the course earlier than the curriculum design. Meanwhile, students with relatively low GPA finish their study longer than eight semesters due to registering many courses after the designated semester. The results can be analyzed further to detect bottleneck and determine possible improvement on the curriculum.

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Keywords: Educational Data Mining; Clustering; Students' Learning Path; Curriculum

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1. Introduction

Educational Process Mining (EPM) is an emerging research area to explore and improve educational process from various educational data. Various data such as massive open online courses (MOOC) and students' academic records from higher education institutions could represent the learning behaviors in many aspects such as time, sequence, context, etc. [1]. By utilizing those educational data, applying various data mining approaches and algorithms can discover hidden knowledge for better decision making for educators, stakeholders, and students.

Curriculum assessment is one of the important step to ensure that a program aligns with the university's mission as well as the education standards from the government. In general, curriculum model provides a guideline for students to consider the learning process during their study period in higher education institutions. The existing work on curriculum assessment relies heavily on a subject-centered approach such as interview and questionnaire. The Chinese University of Hong Kong (CUHK) described the learning outcome to produce graduates who have depth in a specialty, breadth in general knowledge, bilateral/trilingual proficiency, cross-cultural sensitivity, a basket of learning skills, and personal attributes including honesty and integrity. The CUHK has done three phases to achieve the learning outcome in which one of the phases is interviewing students, teachers and alumni. In addition, questionnaire is used to improve teaching and learning environment. However, these approaches neglect students' learning behavior that would give a better insight about curriculum assessment in conformity with the curriculum guideline.

Curriculum mining, as a part of educational process mining, has been proposed as a way to assess the observed students' behavior during the academic period with the curriculum guideline [2]. Using some process mining approaches, curriculum mining attempted to discover the curriculum model from the students' academic data. Although the existing approach could show the effectiveness in the European higher education institutions, there remains questions on the applicability in Indonesia where education curriculum should follow a standard process as an element of a Standards of Education set by the Government described in national review of policies for education by Asian Development Bank [3]. Previous work by Bendatu and Yahya developed an approach, called sequence matching alignment, to check the alignment between the model and the observed behaviour of the students. Although the work presented some interesting findings such as the overall process effectiveness of a curriculum model, the results could not show the potential risks of particular students (e.g., bottleneck, delay, etc.) during the learning process [4]. A risk management approach for curriculum has been proposed by Wong and Lavrencic [5]. However, the risk management model was only applied to evaluate the courses. Brijesh et al uses decision Tree to predict student performance based on student's information [6]. Ben-Zadok et. al. used a case study to analyze the students' learning behaviour to predict the results and to alert the students at risk before their final exams [7]. Finch et al use regression modelling, logistic modelling and network analysis to identify at-risk students and allow for more timely pedagogical interventions [8].

This paper aims to present an aggregate profile clustering approach to categorize students based on their academic profiles. The main idea is to cluster students' learning behaviours which each course is updated by aggregating over matching profiles, called as aggregate profile. For instance, students' academic profiles include courses information such as semester, grade, lecture, etc. Each course with the respective information is mapped into some matching profiles (e.g., match, preceded, succeeded, repetitive, and selective) in accordance to the curriculum guideline. Three steps to obtain aggregate profile is used; match, assign, and aggregate. Each trace (i.e., academic record) will be matched between the observed behaviours and curriculum guideline. The matching result will be a binary value appropriateness factor which is a result of the assign step. Finally, the result is aggregated based on the matching profile. The clustering results could distinguish the learning behaviors according to their performance (i.e., grade point average (GPA) to support students on some factors such as career paths. To verify the clustering results, we analyse each of the clusters using sequence matching alignment [4]. Rather than using the popular approach such as big data, stream data, etc., this study focuses on performance analysis using clustering approach to obtain meaningful knowledge for curriculum alignment. The assessment result can be used to detect bottleneck and provide insight for curriculum development.

The organization of this paper is as follows. Section 2 addresses briefly previous works related to trace clustering. Section 3 describes the methodology used in this study. The implementation and discussion are explained in Section 4. Finally, Section 5 concludes this study.

2. Related work on trace clustering

The availability of data store in organizations database drive the emergence of process mining. While data mining aims to extract knowledge from any types of data, process mining aims to extract process models from event logs. Bose and Aalst explains that an event log corresponds to a set of process instances following a particular business process [9]. A process instance is manifested as a trace whereas a trace is an ordered list of activities invoked by a process instance from the beginning of its execution to the end.

Dekker et al predicting the Electrical Engineering students drop out after the first semester of their studies using university data or even before they enter the study program using pre-university data as well as identifying success-factors specific to the EE program. The result of experiment shows that rather simple classifiers give a useful result with accuracies between 75 and 80%. The predictor of success is the grade of Linear Algebra, Calculus, Networks and mean grade for VWO Science courses [10].

Bendatu and Yahya proposed sequence matching alignment for engineering curriculum using a plugin in process mining framework. Some interesting findings showed that some courses were being taken before, exactly and after the predefined semester. In the conclusion, it was mentioned that many subjects have been taken after the predefined curriculum model [4].

Pechenizkiy et al proposed some of the potential of process mining technique applied to online assessment data. Online assessment data uses the series of the MCQ test organized during mid-term exams at Eindhoven University of Technology using Moodle and Sakai. The demonstration for applicability of process mining uses data collected from two exams. To understand the assessment processes, they use various ProM plug-ins like dotted chart analysis, process discovery (Heuristic Miner and Fuzzy Miner), conformance checking and performance analysis. The result of the analysis shows that almost all students checked their answers and asked for feedback afterwards [1].

Cairns et al developed methods to better understand students' learning habits and the factors influencing their performance. It aims particularly at discovering, analyzing and providing a visual representation of complete educational process. Analysis process used ProM plug-in dotted chart, process modelling (heuristic miner), social network mining and clustering. Social network mining is used to examine and assess interactions between originators, training courses or pedagogical resources, involved in students' training paths. They also proposed a two-step clustering approach to extract the best training paths depending on an employability indicator [11].

Wang and Zaiane applied process mining on curriculum data to provide a means to compare path of successful and less successful students, and presents an opportunity to adjust the requirements or the curriculum. The results can lead students to expected outcome and assist teachers and administrators to understand student's behavior to recommend the correct path to students and enhance curriculum design [2].

Many works have been done to develop process mining algorithms. However, these algorithms have problems to model highly unstructured process and result in spaghetti-like process models that are difficult to comprehend. Trace clustering is an approach that "cluster the traces such that each of the resulting clusters correspond to coherent sets of cases that can each be adequately represented by a process model" [12]. Some trace clustering approaches have been proposed in regard to vector-based profiles [13], generic edit-distance [13, 14], process tree alignment [15], active learning method [16], distance graph model [17], distance metrics between activities of workflow changes and evolution-aware security audits [18], and homogeneous traces to discover simpler models [19]. Among those studies, clustering with vector-based profiles approach is the nearest work to this study. However, primitive profiles in a trace with the same timestamp hinders the clustering approach. This study develops a clustering approach using aggregate profile which is based on the trace profile to categorize students according to the learning paths for each learning period, i.e., semester.

3. Methodology

This study aims to propose a framework to assess students' learning behavior in conformity with curriculum guideline. The framework consists of three major steps; data preparation, curriculum assessment and verification. Data preparation aims to retrieve the data from database, filter and convert the retrieved data for the analysis purpose. Proposed framework can be seen in Fig. 1. Curriculum assessment aims to assess, detect bottleneck and determine future curriculum using the comparison between students' learning behavior and curriculum model.

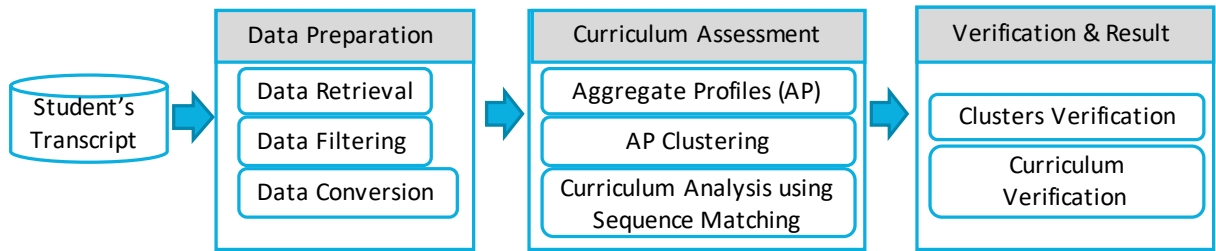


Fig. 1. Analysis framework for curriculum assessment.

3.1. Data preparation

Data preparation aims to obtain the students learning behavior from institution database and prepare the data before the assessment stage. The students' learning behavior can be captured from the students' performance recorded in academic transcripts. In other words, all information related to the students' academic results in the program are retrieved. Since the data includes noise, it is necessary to filter out those noises for the analysis purposes. The noise data such as drop out students and students who took leave during the study are excluded from the database. Finally, the filtered data is converted into a data format that fits for the analysis.

For the analysis, a log that contains a set of cases (i.e. process instances) is required. Each case consists of a sequence of events with various attributes such as activity name, event type, timestamp, resource name, event data, etc. The data retrieved from institution database is students' performance data, which have several attributes such as student ID, course name, course ID, course credit, course grade, high school, etc. In this study, only the relevant attributes are selected including student ID, course name, course ID, course credit, and course grade. It should be noted that course ID and course name refers to the unique property of a course that later be considered as activity. Thus, in the later section, the term course is used to represent the concatenation of those two attributes. The event log can be formulized as follows.

Definition 1. Event Log (L)

An event log (L) is a multiset of traces. Each trace is mapped into a case and consists of a collection of events, which is defined as follows:

E is a set of events and $E = S \times T \times G$ where S is a set of courses, T is a set of timestamps and G is a set of grades.

A set of courses (S) are a collection of course name in the department. Sometime, it can be represented as course identification (course ID). Timestamps (T) is the set of semesters on the study program meanwhile the set of grade (G) designated on the program is illustrated by letter and number such as $\{A, AB, B, BC, C, D, E\}$ and $\{4.0, 3.5, 3.0, 2.5, 2.0, 1.0, 0\}$, respectively.

Case (C) is a set of events for each student where a collection of all cases is an event log (L). In other word, a case represents a student study history. Cases always have a trace, denoted as $\sigma \in E^*$. $C=E^*$ is the set of possible event sequences (traces describing a case).

Table 1 (left) displays a fragment of event log related to educational process from institutional database. The database contains three cases, which refers to student ID $\{1, 2, 3\}$. The timestamp denoted by T1, T2 and T3 refers to Semester 1, Semester 2, and Semester 3, respectively. The set of courses in the database refers to the course name shown in Table 1 (right). Since the current process mining techniques use the basic time granularity, such as second, minute, hour, etc., it has some problems when they are applied to a domain, e.g., education, with high-level time granularity, i.e. a unit of measure of time such as semester, quarter, trimester, etc. Thus, the process mining analysis results in education field are often misleading for the users. To analyse the data, it is necessary to structure the data according to the timestamp. For example, educational process which uses semester holds multi-attributes (e.g., courses, facility, active students, etc.) in a semester.

Definition 2. Projected event log (*PL*)

A projected event log is a multiset of traces which events with the same timestamp are grouped into a set. Projected event (E_t) is defined as a collection of events that happened in the same timestamp. Note that the sequence in projected event is unordered. A projection function $f: L \rightarrow PL$ where $f(C) = \{E_t, t = 1, \dots, n, E_t \subseteq E\}$.

Table 1. Fragment of event log related to educational process (left) projected to multiple events in a timestamp (middle) and course name (right).

Stu.	Timestamp	Event (grade)	Stu.	Timestamp	Events	Course	Course Name	Semester
A	T1	S1 (B)	A	T1	S1, S2	S1	Religion	1
A	T1	S2 (D)	A	T2	S2, S3	S2	Interpersonal Skill	1
A	T2	S2 (BC)	A	T3	S4, S5	S3	English	2
A	T2	S3 (E)	A	T4	S3	S4	Operation System	2
A	T3	S4 (B)	B	T1	S1, S2	S5	Database Management	3
A	T3	S5 (B)	B	T2	S3, S4	S6	Algorithms	3
...			B	T3	S5, S6	S7	Statistics	4
C	T1	S1 (A)	B	T4	S7, S8, S9	S8	Software Engineering	4
C	T1	S2 (AB)	C	T1	S1, S2	S9	Business Planning	4
...			C	T2	S3, S4, S5			

Table 1 (middle) shows a projected fragment of event log related to educational process of students. In the semester T1, student A took two courses, S1, and S2. Each course distinctly and unorderedly shown. In total, student A studied for four semesters with five courses, which are S1, S2, S3, S4 and S5. Two courses S2 and S3 have been repeated as the student fail to satisfy the requirement, e.g. a minimum grade. Student B studies for four semesters and leave of absence on semester S2. Student B took longer time to study and more courses in comparison to student A. The courses (unordered) set for student B are $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$. Looking at the data, it can be seen that student A and B have different behavior during the educational process.

Data conversion aims to convert the students' behaviour data into a format for analysis, i.e., MXML. Table 2 shows the result of data conversion for the analysis purpose.

Table 2. Converted of fragment of a projected event log.

Student	Trace
A	(S1, S2), (S2, S3), (S4, S5), (S3)
B	(S1, S2), (S3, S4), (S5, S6), (S7, S8, S9)
C	(S1, S2), (S3, S4, S5)

3.2. Data preparation

The curriculum assessment considers a projected event log with additional attributes such as students' grade. Before the analysis, we encode the grade to numerical value. For example, the grade A, AB, B, BC, C, D, and E represents the value of 4, 3.5, 3, 2.5, 2, 1, and 0, respectively. The numerical value is used to measure the student's performance, i.e. Grade Point Average (GPA). Moreover, it is necessary to recode the curriculum guideline. Curriculum guideline refers to the prior design of the curriculum. The curriculum guideline (*CG*) can be formulized as follow. $CG = \{cg_{ir} \mid i = 1, \dots, n; r = 1, \dots, m\}$ where i refers to the index of i -th semester and n is the maximum semester allowed, r refers to the r -th index of course and m is the total courses. The curriculum guideline is converted into a data structure to match with the projected event log. Although in many curriculum guidelines include prerequisite information, this study limits the study scope unto the appropriateness factors and reserves the sequential learning path to future work. Note that the index of courses in the curriculum guideline is based on the lexicography of the course name and the course identification.

For the purpose of clustering, the data is aggregated and processed according to the appropriateness factors based

on the type of courses (i.e., compulsory, general, and selective) with three steps; match, assign, and aggregate. This “match” step aims to measure the alignment of student’s study path with the curriculum. The way to measure the alignment is using sequence matching alignment. The measure will check the semester of course taken by the student and the semester of course designed in the curriculum. To check the semester of course taken by the student, we need to analyse the trace of the event log. As aforementioned, the trace based on event log is a sequential data which represent the timestamp of the behavior. For example, the first set of courses represent the courses taken on the first semester. Let σ be a trace of a student. By referring to the trace on the event log, student’s learning behavior can be denoted as σ_{ir} where i refers to the index of semester and r refers to the index of courses. In other word, the trace is segmented according to the index of semester. Note that the index of the course is ordered based on either lexicography or instructor opinions. Since there might be retaking courses, we eliminate the retaking courses and denote it as f -segmented-trace. For example, the f -trace of student A is $\langle (S1, S2), (S3), (S4, S5) \rangle$. Hence, the matching analysis is measured as follow.

$$m_s(cg_{ir}, \sigma_{ir}) = \begin{cases} 1 & \text{where } cg_{ir} = 0 \wedge \sigma_{jr} = 1 \wedge i < j \\ 0 & \text{where } cg_{ir} = 1 \wedge \sigma_{jr} = 1 \wedge i = j \\ -1 & \text{where } cg_{ir} = 0 \wedge \sigma_{jr} = 1 \wedge i > j \end{cases} \quad (1)$$

where $0 < i, j < n \wedge \forall r$. For example, Table 1 (right) have included the semester information from the curriculum guideline. If a student took "Interpersonal Skill" in semester 1, as arranged in the curriculum design, then it is matched ($ms = 0$). Otherwise, it is unmatched. The step “assign” aims to measure the number of courses which does not match the curriculum design and assign them in specific categories. There are three categories for these assignments; preceding course, succeeding course and repetition course. Preceding course aims to measure the number of courses preceded the designated semester in the curriculum design meanwhile succeeding course aims to measure the number of courses succeeded the designated semester in the curriculum design. Intuitively, students who intend to graduate fast will consider taking courses earlier than the designed semester. On the other hand, some students may carefully choose courses and take those later than the design. For example, student A took "Operation System" in Semester 3 represents as succeeding course ($ms = 1$) and student C took "Database Management" in Semester 2 represents as preceding course ($ms = -1$).

Some students may retake the courses due to the bad grades and constraints designed by the curriculum administrator. Hence, the repetition course aims to measure the number of courses repeated by students during the study period. In addition, we include selective courses that aims to measure the number of courses that are chosen by the students as their future expertise. Since the selective courses are different for each semester, this analysis attempts only measure the number instead of matching alignment. To detect the selective courses, we measure the compulsory courses. Compulsory courses mean that a set of courses need to be taken during the study period. In other word, the frequency of a course should equal to the number of students. Otherwise, it is a selective course.

Finally, the “aggregate” phase aims to group the appropriated factors by summing all the values based on the semester and students ID. This step will result a database for clustering purpose. Table 3 shows the result of aggregation.

Table 3. Structuring the data (aggregate profile) for clustering purpose.

Student ID	Match	Preceding	Succeeding	Repetition	Selective
A	4	0	1	2	0
B	9	0	0	0	0
C	4	1	0	0	0

Clustering is a method to group objects in such a way that similar objects are placed in a group and different objects are placed in another group. In this research, student clusters analysis was used to determine the composition of student learning profiles. One of the popular approach in clustering is k-Means. K-Means clustering is a method to result a k cluster, which k is defined by user. The group is selected by finding the smallest sum of squared error (SSE) in comparison with mean.

Given a set of students $(c_1, c_2, \dots, c_n)$, where each student has d -dimensional integer vector, k -means clustering aims to partition the n observations into k ($\leq n$) sets $C = \{C_1, C_2, \dots, C_k\}$ to minimize the within-cluster sum of squares error. The objective of aggregate profile (AP) clustering is denoted as follow.

$$\operatorname{argmin}_C \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|^2 \quad (2)$$

where μ_i is the mean of points in C_i .

In the curriculum analysis, the result of sequence matching and students' clusters is shown in a form of table. The table displays the courses taken by students in each semester. The column represents the semester and the row represents the course lists taken by students. To evaluate the alignment of the curriculum, we develop three measures; course matching rate, semester matching rate, and overall matching rate. The course matching rate (cm_i) is measured as the number of students (c_i) who took the i -th course on the designated semester over total students (the number of cases, denoted as $|C|$). The semester matching rate (sm_j) is measured as the summation of matching rate of course in the j -th semester over the number of courses in that semester (k). This measure is to show the effectiveness of courses in the curriculum guideline in the respective semester. The maximum value is 1, which means that the courses in j -th semester are in the right period. The overall matching rate (om) is measured as the summation of course matching rate over the number of courses designated in the curriculum design. This measure is to show the effectiveness of all courses in the curriculum guideline of the department. The maximum value is 1 which means that all students took the courses exactly the same as the curriculum guideline. Reader can refer to Bendatu and Yahya [4] for further details.

4. Implementation

This section aims to show the implementation results and discusses some relevant issues of the current work. First, the data used in this study is described. Second, the tool used for curriculum assessment is explained. This study applies a case study from an Information Systems Curriculum of a public university in Surabaya, Indonesia. The data used in this study consists of students who studied using curriculum 2009 as the model. The data which consists of noises has been filtered. For example, data which consists transferred and drop-out students has been ignored. After the filtering process, there were 182 students' performance data for the assessment step. The filtered data is further converted to MXML for curriculum assessment.

The implementation of this approach is using Java programming language. It is implemented as a plugin in process mining tool, called ProM. The WEKA library is embedded in the tool for the student clusters. In process mining, event log is ordered based on the timestamp. Since the timestamp of this students' performance data has many duplication, we distinct the data into three columns; caseID, timestamp, activity. In the column of activity, it includes the set of courses taken by a student on a timestamp. Since there can be a lot of activities in a set, then this set is lexicographically ordered. For clustering, we use $k=3$ on the k -means approach. The result of student clusters categorizes the students into three groups. Descriptive statistics for each cluster is shown in Table 4.

Table 4. Descriptive statistics for each cluster.

	# of students	Overall Matching	Overall Avg. GPA	Max. GPA	Min GPA	Stdev GPA	Max Semester
Cluster 1	114	0.686	3.389	3.795	2.864	0.196	10
Cluster 2	48	0.676	3.564	3.875	3.256	0.140	8
Cluster 3	20	0.662	3.058	3.392	2.802	0.202	10

Table 4 showed overall characteristics of each cluster. Cluster 2 consists of high performers with the highest average GPA compared to other clusters and the maximum length of study of 8 semesters. Cluster 3, on the other hand, consists students who had the lowest average GPA and the maximum length of study of 10 semesters. Cluster 1 which has the highest number of students among the clusters can be seen as the average in terms of the performance in comparison to cluster 2 and cluster 3. It can be seen from the Overall Average GPA that the value of cluster 1 is

higher than cluster 3. However, some of the students (in Cluster 1) took 10 semesters to finish their study.

After the clustering analysis, we perform curriculum analysis using sequence matching method. The developed tool for sequence matching can show the compulsory, selective, and retaking courses. The result from cluster 1 shows that semester 6, 7 and 8 are the period where most of students attempt to either took selective course or retook the course to improve their score. The result of cluster 2 shows that semester 5, 6 and 7 is the period where most of students attempt to either took selective course or retook the course to improve their score.

The result also shows that there are only 2 courses taken by students in semester 9 in Cluster 1. However, in Cluster 3, there are up to 10 courses that should be taken in earlier semesters according to curriculum guideline at Semester 9. This indicates that the students in Cluster 3 retook those courses later than any other courses.

Example of the fragment of matching analysis for Cluster 2 can be seen in Fig. 2. It has information on each column for semester, courses name, FBM (Frequency Before Matching), RBM (Relative Before Matching), FM (Frequency Matching), RM (Relative Matching), FAM (Frequency After Matching), and RAM (Relative After Matching). The result shows that all courses in semester 1 have RM=1. It means all the students took the courses exactly the same as the curriculum guideline since all of these courses are compulsory.

After the first semester, the students have the right to choose the courses. In overall, the RM value which is greater than 0.5 for the three clusters mean that most of the students took the courses the same as the period at the curriculum guideline. It is shown based on the result on semester 2 until the last semester. Some courses have RBM and RAM value representing courses that student taken before and after the designated semester in curriculum guide. Matching analysis for cluster 3, “Language” have value of FBM=1, RBM = 0.05, FM=12, RM=0.6, FAM=7 and RAM=0.35. It means 12 from 20 students in this cluster (60%) took this course exactly like in the curriculum, 7 students (35%) took this course after semester defined in the curriculum and only 1 student (5%) took this course before the semester defined in the curriculum.

The summary of matching analysis result can be seen in Table 5. This result shows the different learning behaviour among the clusters. Interestingly, it also relates to the students’ performance. Cluster 2 which consists of high performers have the highest Average RBM which means the students in this cluster take courses before the designated semesters. Cluster 2 also has the lowest RAM compared to other clusters which means the member of the clusters do not take courses later than the indicated semester in the curriculum guide. In comparison, cluster 3 with low performing students have most courses that had been taken after the defined semester. Interestingly, cluster 3 also have a relatively high RBM, which mean they also tend to take course before the defined semester. This show the variety of learning behaviour of the students in this clusters.

Matching Analysis View

Semester	Course	FBM	RBM	FM	RM	FAM	RAM
1	Algorithm and Programming 1	0	0.0	48	1.0	0	0.0
1	Interpersonal Skill	0	0.0	48	1.0	0	0.0
1	Discrete Mathematics	0	0.0	48	1.0	0	0.0
1	Introduction To Information & Comm. Technology	0	0.0	48	1.0	0	0.0
1	Business System Functionality 1	0	0.0	48	1.0	0	0.0
2	Algorithm and Programming 2	0	0.0	48	1.0	0	0.0
2	English	0	0.0	48	1.0	0	0.0
2	Management and Organization	0	0.0	48	1.0	0	0.0
2	Introduction to Information System	0	0.0	48	1.0	0	0.0
2	Business System Functionality 2	0	0.0	48	1.0	0	0.0
2	Operating System	0	0.0	48	1.0	0	0.0

Fig. 2. Example of matching analysis view for cluster 2.

There are 7, 4, and 2 courses in cluster 1, 2, and 3, respectively, with RBM value greater than 0.5. The courses taken before the designated semesters are “Statistics”, “Introduction to Technopreneurship”, “Civics”, “Information Technology Investment Management”, “Professional Ethics”, “Capita Selecta” and “Information Technology Monitoring and Evaluation”. This is possible as there are no pre-requisites on these courses. Hence most students would like to take it earlier as they still had extra credits in the semester. This analysis may indicate that the courses can be put in earlier semesters in the curriculum.

Table 5. Summary of matching analysis.

	RBM	RAM	Average RBM	Average RM	Average RAM
Cluster 1	20	11	0.153	0.686	0.008
Cluster 2	21	4	0.174	0.676	0.004
Cluster 3	16	27	0.090	0.662	0.103

5. Conclusion

This study proposes a methodology to assess the curriculum alignment with aggregate profile clustering and sequence matching. The existing plugin in ProM (i.e., sequence matching) has been extended to assess the curriculum as well as cluster the students according to the alignment with the curriculum guideline. The methodology had been tested to analyze the curriculum using Information Students' data. Results from the implementation showed that the IS students can be grouped into three clusters that have diverse characteristics with respect to their performance and learning behaviour. It was found that students who perform highly in terms of average GPA tend to take the course earlier than the curriculum guideline and a few courses was taken after the designated semesters. On the other hand, low performance students not only took a lot of courses early, but also took many courses after the designated semester. Therefore, the low performance students finished their study longer than the average, which is set as 8 semesters. This result can be used to improve the curriculum further.

There are some issues opened for future work. Large data set from batches that experience the same curriculum guideline will give more comprehensive result. Moreover, the existing analysis can also be used to find the bottleneck of students' learning process. Another important question for curriculum assessment is toward the alignment between the observed behaviour with the curriculum guideline when there are some changes on the curriculum contents, later considered as a concept drift. Using segmented clustering approach, there might be some interesting results to address concept drift in educational process mining. Other issues are related to auxiliary attributes such as gender and students' historical background. The relationship between the learning process and those attributes could be an input factor for curriculum development.

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